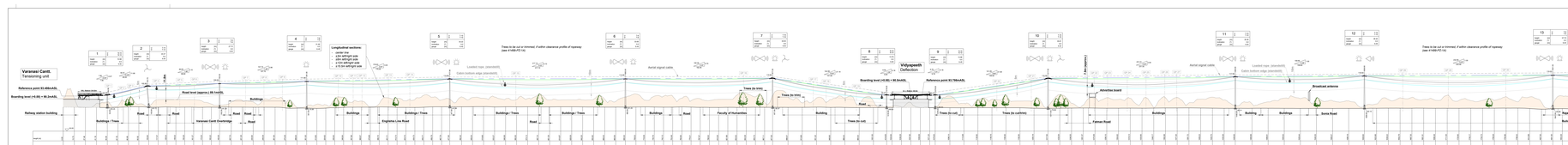

Varanasi Cantt.
Tensioning unit
Reference point 93.486mASL
Boarding level (+0.00) = 90.2mASL
XXL-Station 28.52m
Railway station building
Buildings / Trees
Road
Varanasi Cantt Overbridge
Road level (approx.) 89.1mASL
Buildings
11.8m
Longitudinal sections:
- center line
- ±3m left/right side
- ±6m left/right side
- ±12m left/right side
- ±15.5m left/right side
Englishia Line Road
Loaded rope (standstill)
Cabin bottom edge (standstill)
Trees to be cut or trimmed, if within clearance profile of ropeway (see 41499-PZ-1A)
Aerial signal cable
Faculty of Humanities
Trees (to trim)
1 U N-14 D N-14 height [m] 10.98 inclination [°] 0.0 gauge [m] 6.45
2 U T-14 D T-14 height [m] 22.27 inclination [°] 5.0 gauge [m] 6.45
3 U T-6 D T-6 height [m] 27.72 inclination [°] 0.0 gauge [m] 6.45
4 U T-6 D T-6 height [m] 31.82 inclination [°] 0.0 gauge [m] 6.45
5 U T-10 D T-10 height [m] 33.22 inclination [°] 0.0 gauge [m] 6.45
6 U T-8 D T-8 height [m] 33.65 inclination [°] 0.0 gauge [m] 6.45
7 U T-16 D T-16 height [m] 34.63 inclination [°] 0.0 gauge [m] 6.45
8 U N-10 D N-10 height [m] 13.79 inclination [°] 0.0 gauge [m] 6.45
9 U N-10 D N-10 height [m] 13.88 inclination [°] 0.0 gauge [m] 6.45
10 U T-16 D T-16 height [m] 35.81 inclination [°] 0.0 gauge [m] 6.45
11 U T-10 D T-10 height [m] 35.30 inclination [°] 0.0 gauge [m] 6.45
12 U T-10 D T-10 height [m] 36.40 inclination [°] 0.0 gauge [m] 6.45
13 U T-8 D T-8 height [m] 37.70 inclination [°] 0.0 gauge [m] 6.45
14 U T-8 D T-8 height [m] 40.96 inclination [°] 0.0 gauge [m] 6.45
15 U T-14 D T-14 height [m] 46.71 inclination [°] 0.0 gauge [m] 6.45
16 U T-16 D T-16 height [m] 32.98 inclination [°] -5.0 gauge [m] 6.45
17 U N-12 D N-12 height [m] 14.70 inclination [°] -5.0 gauge [m] 6.45
18 U N-12 D N-12 height [m] 13.01 inclination [°] 0.0 gauge [m] 6.45
Vidyapeeth
Deflection
Boarding level (+0.00) = 90.5mASL
Reference point 93.786mASL
2x L-Station 46.6m
Trees (to cut)
Building
Trees (to cut/trim)
7.9m (approx.)
Advertise board
Fatman Road
Broadcast antenna
Sonia Road
Raja Sir Motichand Road
Broadcast antenna (to relocate)
Advertise board (to relocate)
Max. height 44.1m
12.6m
Luxa Road
Rath Yatra
Drive unit
Reference point 91.786mASL
Boarding level (+0.00) = 88.5mASL
XXL-Station 28.52m
Section 2
length [m]
altitude [m ASL]
Corridor ropeway
Corridor fire protection
18.7m

Ropeway Calculation of Bartholet

system data	
Type	GBK10P
drive station	Top Station
tension station	Bottom Station
capacity of transporting [people/h]	3000.0
return transport [%]	100
sense of rotation	clockwise
rope speed [m/s]	6.0
station speed [m/s]	0.28

geometrical data	
rope height on top [m ASL]	92.6
rope height on bottom [m ASL]	94.3
vertical rise [m]	1.7
horizontal length [m]	2295.9
inclined length [m]	2295.9
medium rope climb [°]	0.0
maximum rope climb [°]	23.8

Vehicles	
number	90
time spacing [s]	11.9
distance between vehicles [m]	71.7
travel time [min]	8.96

engine	
power continuous [kW]	560.5
power start [kW]	788.1

rope	
diameter [mm]	52.0

C	2.0.0.9	25.05.23 GROM	25.05.23 BEAN
Rev.	Prgversion	calculated	reviewed

Rope cordline
Empty rope +25%
Loaded rope +25%
Vehicle underside +25%

▨ Filling
▨ Excavation
Spotlights
Camera
Speakers
Anemometer
Obstruction lights

Topographical data provided by: *Vishwa Samudra Engineering Private Ltd. (January 2023)*

INDEX	REVISION	DRAWN		REVIEWED	
D	Loudpeakers and aviation obstruction lights	14.12.23	HAUA	15.12.23	FO
A	-	17.02.23	GROM	17.02.23	BEAN

Longitudinal profile - Section 1

GBK10 Varanasi Cantt-Vidyapeeth-Rath Yatra
Varanasi (Uttar Pradesh), India

SCALE
1:1000

BARTHOLET

No.	of	DRAWING-NUMBER	SIZE
1	1	41499-LP-1D	A4

H:\100 Projekte\40000 Seilbahnen\41499 GBK10 Varanasi (IND)\TS00 - Gesamtanlage\02 - Seilrechnung - Längenprofil\41499-SR-1D-Arbeit\41499-LP-1D.dwg

The detailed engineering drawings are to be used for the execution.
The foundations will be executed according to the information of the civil engineer.

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